A smoothed Frobenius determinant bound for the tenth discriminant

Version 0.6.0: $q < 570$

Research note prepared with Codex

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Abstract

We obtain $q < 570$ for an imaginary quadratic field of class number one. The new argument excludes $570 \le q \le 2000$; the previously audited v5 result supplies the range above 2000. The proof retains a smoothed prime sum already present in the determinant valuation, uses it directly in an acyclic factorial refinement, and adds compatible higher Frobenius factors. A pointwise inequality couples derivative orders with interpolation radii. A positive kernel formed from a squared polynomial completes the comparison with the quadratic norm series. All determinant parameters are fixed, apart from the scale and its auxiliary-prime interval. The numerical certificate uses rational intervals and Python's standard library. The external number-theoretic inputs are quadratic ideal theory and explicit ordinary-prime estimates of Rosser and Schoenfeld. No quasi-Riemann hypothesis or zero-free theorem for quadratic Dirichlet functions is used.

1 Scope, provenance, and the result

For a negative fundamental discriminant D, put $q = |D|$ and $K = \mathbb{Q}(\sqrt{D})$. We address Tao's request for tenth-discriminant bounds obtained without the stronger zero-location machinery discussed in the comments to his challenge [1, 2].

Theorem 1. *If $h(K) = 1$, then $q < 570$.*

The sole earlier-version input is the independently audited implication $h(K) = 1 \Rightarrow q < 2000$ from v5 [10]. The entire new exclusion of $[570, 2000]$ is proved here, relative to the classical prime estimates stated below. None of the intermediate unpublished 1600, 1100, or 1050 drafts is a dependency. Discriminants in the new interval are not enumerated to obtain the analytic cutoff.

The framework is Green's derivative determinant [3], with van Dijk's auxiliary inert prime for nonvanishing [4]. Green's comment also records a related unpublished argument of Jacob Pescador [5]; we have not used that unpublished text. Pratt proposed combining auxiliary inert primes from quadratic norm counts, Green's stronger divisibility, and disk interpolation [6]. Kerr's suggestion to retain higher Frobenius relations, including the ninth-power relation at 3, also informed the valuation refinements [7]. The smoothed comparison, direct factorial refinement, compatible residual factors, and joint archimedean estimate below were developed in this AI-assisted research session. No priority claim is made for these refinements. The Poisson expansion of the quadratic norm series is classical; Ross and Tao discussed its usefulness in this challenge [8, 2]. We use it in the half-plane $s > 1$, with an explicit error bound for the signed kernel comparison.

This v6 draft has not yet had an independent mathematical audit. Exact computational checks are distinct from an audit of the general argument and from formal verification. Earlier versions retain their own review records; those reviews do not cover this draft.

2 Elementary consequences of class number one

Assume henceforth that $h(K) = 1$ and $570 \le q \le 2000$. Every nonrational algebraic integer has norm at least $q/4$. A noninert rational prime has an integral ideal of norm p; under class number one its generator is nonrational. Consequently

$$p < q/4 \quad \Longrightarrow \quad p \text{ is inert.} \tag{1}$$

In particular q is an odd prime: an even discriminant has a small ramified prime 2, and a composite odd discriminant has a ramified prime at most $\sqrt{q} < q/4$. Inertness of 2 gives $q \equiv 3 \pmod 8$. The only units are ± 1, and

$$\mathcal{O}_K = \mathbb{Z} + \mathbb{Z}\omega, \qquad \omega = (1 + i\sqrt{q})/2, \qquad Q_q(a,b) = |a + b\omega|^2 = (a + b/2)^2 + qb^2/4.$$

Write

$$v = \sqrt{q}/2, \qquad \rho = \tfrac{1}{2}\sqrt{1 + v^2}.$$

The lattice has covolume v and a rectangular tiling of covering radius ρ. Its disk count $N(r)$ therefore satisfies

$$\frac{\pi \max(r - \rho, 0)^2}{v} \le N(r) \le \frac{\pi (r + \rho)^2}{v}. \tag{2}$$

Every principal ideal of norm m is a rotation and dilation by $\sqrt{m}$ of this lattice.

3 The ordinary-prime inputs

Put $\vartheta(x) = \sum_{p \le x} \log p$ and $A(x) = \sum_{p \le x} \log p/p$. We use Rosser–Schoenfeld [9], with $c_0 = 1.01624$:

$$\vartheta(x) < c_0 x, \tag{3}$$

$$x\left(1 - \frac{1}{2\log x}\right) < \vartheta(x) \quad (x \ge 563), \tag{4}$$

$$\vartheta(x) < x\left(1 + \frac{1}{2\log x}\right) \quad (x > 1), \tag{5}$$

$$A(x) > \log x - 1.334 - \frac{1}{2\log x} \quad (x \ge 319). \tag{6}$$

The last constant follows from their Theorem 6 and the explicit prime sum at 319: the certificate checks $\log 319 - A(319) + 1/(2\log 319) < 1.334$.

For sharper bounds at our scales, Table I on printed page 90 gives pairs (b, ϵ_b) such that $(1 - \epsilon_b)x < \psi(x) < (1 + \epsilon_b)x$ for $x \ge e^b$. The corollary to their Theorem 14 gives $0 \le \psi(x) - \vartheta(x) < 1.02\sqrt{x} + 3x^{1/3}$. Thus, uniformly for $\log x \ge m \ge b$,

$$1 - \epsilon_b - 1.02e^{-m/2} - 3e^{-2m/3} < \frac{\vartheta(x)}{x} < 1 + \epsilon_b. \tag{7}$$

Let $c(m), d(m)$ be any certified upper and lower bounds obtained by taking the best of these inequalities and (3)–(5). The table entries used are included as exact decimals in `parameters.json`.

We also use $A(R) < 2\log R$ at our scales. This follows without a further prime estimate: $\log(R!) \ge RA(R) - \vartheta(R)$, so $A(R) \le \log R + c_0 < 2\log R$.

4 A sieve providing the auxiliary inert prime

Fix one band $Q_0 \le q \le Q_1$. Form the fixed, finite list of pairs

$$b > 0, \quad 2a + b > 0, \quad \gcd(a, b) = 1, \quad Q_{Q_1}(a, b) < (Q_0/4)^2, \tag{8}$$

ordered by $(Q_{Q_1}(a, b), a, b)$. Write $p_j(q) = Q_q(a_j, b_j)$, $p_j^- = Q_{Q_0}(a_j, b_j)$, and $p_j^+ = Q_{Q_1}(a_j, b_j)$.

Each $p_j(q)$ is prime. A composite norm below $(q/4)^2$ has a prime factor below $q/4$; inertness would force that prime to divide both coordinates. It is split rather than ramified: generators of the unique prime ideal above q have zero real part. These primes are distinct, because the four generators of a split prime norm are $\alpha, -\alpha, \bar{\alpha}, -\bar{\alpha}$, and the chosen quadrant contains exactly one of them.

Choose the first n such split primes, and the first h rational primes $r > 7$ with $r < Q_0/4$. Sieve by the two prime ideals above each selected p_j, and by each ideal $r\mathcal{O}_K$. Their norms are p_j, p_j, r^2, and they are pairwise coprime. Also impose the exact condition $\gcd(a, b, 210) = 1$.

If an element belongs to k of the selected ideals, then $1 - k + \binom{k}{2}$ is an upper bound for the indicator that it belongs to none. Apply this second Bonferroni inequality, multiplied by the exact 210 indicator, to the annulus $U < |\alpha|^2 \le \kappa U$. A split prime in this interval has four generators. Assume $U > Q_1$ and $U > \max_{j \le n} p_j^+$, so the sieving primes themselves are not counted.

For an ideal of norm m, the disk estimate gives an annulus count with area term $\pi(\kappa - 1)U/(vm)$, boundary error at most $2\pi\rho(1 + \sqrt{\kappa})\sqrt{U}/(v\sqrt{m})$, and constant error at most $2\pi\rho^2/v$. The 210 expansion has

$$\sum_{d|210} \frac{\mu(d)}{d^2} = \frac{768}{1225}, \qquad \sum_{d|210} \frac{|\mu(d)|}{d} = \frac{96}{35}, \qquad \sum_{d|210} |\mu(d)| = 16.$$

Define

$$u = 2\sum_{j \le n} (p_j^+)^{-1} + \sum_r r^{-2}, \qquad u_2 = 2\sum_{j \le n} (p_j^+)^{-2} + \sum_r r^{-4},$$
$$z = 2\sum_{j \le n} (p_j^-)^{-1/2} + \sum_r r^{-1}, \qquad z_2 = 2\sum_{j \le n} (p_j^-)^{-1} + \sum_r r^{-2}, \qquad m_0 = 2n + h.$$

The certificate checks that the analogous u at the minus endpoints is less than 1. The area polynomial then decreases in every reciprocal ideal norm, since its partial derivative is -1 plus the sum of the other reciprocals. Therefore uniform upper bounds for the number of split primes in $(U, \kappa U]$ are

$$A_*(\kappa - 1)U + B_*(1 + \sqrt{\kappa})\sqrt{U} + C_*, \tag{9}$$

where $v_- = \sqrt{Q_0}/2$, $v_+ = \sqrt{Q_1}/2$, and

$$A_* = \frac{\pi}{4v_-}\frac{768}{1225}\left(1 - u + \frac{u^2 - u_2}{2}\right),$$
$$B_* = \frac{\pi}{4}\sqrt{1 + v_-^{-2}}\frac{96}{35}\left(1 + z + \frac{z^2 - z_2}{2}\right),$$
$$C_* = 2\pi(v_+ + v_+^{-1})\left(1 + \frac{m_0(m_0 + 1)}{2}\right).$$

For a parameter row, put $M = 2^a$, and let $2^a \le U \le 2^b$. Set $d = d(\log M)$, $c = c(\log M)$. An inert prime exists in $(U, \kappa U]$ if

$$G(x) = \kappa d - c - (x + \log\kappa)\{A_*(\kappa - 1) + B_*(1 + \sqrt{\kappa})e^{-x/2} + C_* e^{-x}\} > 0, \quad x = \log U. \tag{10}$$

Indeed, the total prime logarithmic weight exceeds the possible split weight, and the ramified prime is below the interval. Since $x + \log\kappa > 4$, the function G is concave. It suffices to check its two endpoints. These are exact rational interval checks in the certificate; all κ, n, h, a, b are in the parameter file.

5 Nonvanishing with the prime as monomial width

Take an inert prime $p_0 \in (U, \kappa U]$, and set $R = p_0$. Throughout the new proof use the fixed constants

$$t = \frac{3}{64}, \qquad C = \frac{27}{64}, \qquad S = \lfloor tR \rfloor, \qquad J = \lceil R/9 \rceil, \qquad T = \sqrt{CRv/\pi} + \rho. \tag{11}$$

Use all n lattice points z_i in the disk of radius T. Then $n \geq CR$, so $nJ \geq tR^2 \geq RS$. The matrix has rows $0 \leq r < R, 0 \leq s < S$, columns (i, k), $0 \leq k < J$, and entries

$$\partial_z^k (z^r w^s)\big|_{(z_i, \bar{z}_i)}.$$

Lemma 2. *If $2T < R$ and $J \leq R$, this matrix has full row rank.*

Proof. Reduce modulo $p_0\mathcal{O}_K$, a field of p_0^2 elements. The points are distinct: a nonzero element of $p_0\mathcal{O}_K$ has modulus at least p_0, whereas two points in the disk differ by less than $2T$. Frobenius gives $\bar{z}_i = z_i^{p_0}$. If a nonzero row combination vanished, substitute $w = z^{p_0}$ in its polynomial. The exponents $r + p_0 s$ are distinct, so the result is a nonzero polynomial of degree at most $RS - 1$. For $k < J \leq p_0$, its ordinary derivatives agree with the specified z-derivatives in characteristic p_0. Each node is thus a zero of multiplicity at least J, since $k! \neq 0$ for $k < p_0$. This gives at least $nJ \geq RS$ zeros, a contradiction. Full rank after reduction implies full rank in characteristic zero. □

Fix a nonzero RS-square minor $\mathcal{D}$, and put $N = RS$ and $\Delta = |\mathcal{D}|^2 \in \mathbb{Z}_{>0}$. Different triangular row changes below may be used for different primes: they leave this same determinant unchanged.

6 The smoothed prime sum already in the valuation

For $0 < u \leq 1$, put

$$f(u) = 2au - a(a+1)u^2, \qquad a = \lfloor 1/u \rfloor; \qquad f(u) = 0 \ (u > 1), \quad f(0) = 1. \tag{12}$$

This is continuous, nonnegative, nonincreasing, and at most 1. Also

$$f(u) = 2u \sum_{m \geq 1} (1 - mu)_+, \qquad \frac{2}{R^2} \sum_{r=0}^{R-1} \left\lfloor \frac{r}{p} \right\rfloor = \frac{f(p/R)}{p}. \tag{13}$$

Define the noninert sum

$$\mathcal{T}(X) = \sum_{p \text{ noninert}} \frac{\log p}{p} f(p/X). \tag{14}$$

It is nonnegative and nondecreasing in X, and is at most the ordinary cumulative noninert sum.

Green's triangular change, using $a_{r,s} = \min(\lfloor r/p \rfloor, S - 1 - s)$, replaces $z^r w^s$ by $z^{r-pa_{r,s}}(z^p - w)^{a_{r,s}} w^s$. For inert p, every ordinary derivative of $z^p - w$ is divisible by p, so

$$v_p(\Delta) \geq 2 \sum_{r,s} a_{r,s}.$$

Write this normalized valuation as $e(R,S,p)$. Counting layers gives

$$e(R,S,p)=\frac{2}{R^2S}\sum_{h=1}^{S-1}(S-h)(R-ph)_+. \tag{15}$$

For $p(S-1)\le R-1$, this equals $(S-1)/R-p(S^2-1)/(3R^2)$. In general,

$$e(R,S,p)\ge \frac{f(p/R)}{p}-\frac{R-1}{2RSp}-\frac{(R-1)(2R-1)}{6RSp^2}. \tag{16}$$

Indeed, sum $\sum_{s<S}\min(k,S-1-s)\ge Sk-k(k+1)/2$, then use $k(k+1)\le r^2/p^2+r/p$ and (13).

The main term in the inert-prime sum in (16) is exactly

$$\sum_p \frac{\log p}{p}f(p/R)-\mathcal{T}(R).$$

Keeping $\mathcal{T}$, instead of replacing it by the larger ordinary cumulative sum, is an important improvement.

6.1 The all-prime floor remainder

If $R=ap+b$, the exact floor identity gives

$$\frac{f(p/R)}{p}=\frac{1}{p}-\frac{1}{R}+\frac{1}{R}\delta_a(p/R),\qquad \delta_a(u)=2a+1-u^{-1}-a(a+1)u.$$

On $[1/(a+1),1/a]$, δ_a is nonnegative, vanishes at the endpoints, and has one maximum. Partial summation with $d_*x\le\vartheta(x)\le c_*x$, valid for $x\ge M/33$, gives

$$G_{\mathrm{floor}}=\max(0,Bd_*-Ac_*),\quad A=\frac{\log 33-H_{33}+1}{2},\quad B=\frac{H_{32}-\log 33}{2}. \tag{17}$$

Here $H_n=\sum_{j\le n}1/j$. The increasing and decreasing pieces contribute integrals $\frac{1}{2}\log(1+1/a)-\frac{1}{2(a+1)}$ and $\frac{1}{2a}-\frac{1}{2}\log(1+1/a)$, respectively; sum for $1\le a\le 32$. Thus

$$\sum_p\frac{\log p}{p}f(p/R)\ge A(R)-c(\log M)+G_{\mathrm{floor}}\qquad (R\ge M). \tag{18}$$

6.2 Removing known noninert primes from the quadratic loss

Put $H=\lfloor 1/t\rfloor=21$,

$$a_H=\sum_{p\le H}\frac{\log p}{p},\quad \theta_H=\sum_{p\le H}\log p,\quad W_H=\sum_{p\le H}\log p\,(t-pt^2/3).$$

Let B_H be a certified upper bound for $\sum_{p>H}\log p/p^2$. Specifically, with $P=10000$, partial summation and (3) give

$$B_H=\sum_{H<p\le P}\frac{\log p}{p^2}+\frac{2c_0}{P}-\frac{\vartheta(P)}{P^2}>\sum_{p>H}\frac{\log p}{p^2}.$$

With all pairs in (8), define

$$K_2 = \sum_j \frac{\log p_j^+}{(p_j^+)^2} + \frac{\log Q_1}{Q_1^2}. \tag{19}$$

The summands decrease with the prime, so the inert quadratic tail is at most $B_H - K_2$. Combining (16), (18), the exact small-prime formula, and $A(R) < 2\log R$, gives

$$\begin{aligned}\frac{\log \Delta}{NR} \geq A(R) - c(\log M) + G_{\text{floor}} - \mathcal{T}(R) - a_H + W_H \\ - \frac{B_H - K_2}{3(t-1/M)} - \frac{2\theta_H}{M} - \frac{\log M}{(t-1/M)M}.\end{aligned} \tag{20}$$

The next two sections give compatible additional gains.

7 Compatible higher Frobenius factors

At an inert prime use

$$G = z^p - w, \qquad H_1 = z^{p^2} - z, \qquad H_2 = w^{p^2} - w.$$

For each row let

$$a = \min(\lfloor r/p \rfloor, S - 1 - s), \quad c = \lfloor (r - pa)/p^2 \rfloor, \quad b = \lfloor s/p^2 \rfloor,$$

and replace the row polynomial by

$$z^{r-pa-p^2c} G^a H_1^c \, w^{s-p^2b} H_2^b. \tag{21}$$

Its leading term is $z^r w^s$; all other terms have smaller lexicographic index and remain in the rectangle, since their z-degree is at most r and their w-degree at most $s + a \leq S - 1$.

All three factors have values divisible by p. A Hasse derivative can lose a factor G only at order p, and a factor H_1 only at order 1 or p^2. At total order k, the number of lost factors is therefore at most

$$L = \min(k, c) + \left\lfloor \frac{k - \min(k, c)}{p} \right\rfloor.$$

Ordinary differentiation supplies $k!$. The elementary inequality

$$c + \lfloor k/p \rfloor - L \geq \lfloor c/p \rfloor$$

holds both for $k \geq c$ and $k < c$. Consequently the entry valuation is at least

$$a + b + \lfloor c/p \rfloor + \sum_{e \geq 2} \lfloor k/p^e \rfloor. \tag{22}$$

The additional residual row factor is $\lfloor (r - pa)/p^3 \rfloor$. Moreover, beyond this bound one may retain

$$\max\left(0, (1 - 1/p)(c - k) - 1\right). \tag{23}$$

For $k < c$, this follows by bounding $\lfloor k/p \rfloor$ below by $k/p - 1$; for $k \geq c$, (22) suffices.

7.1 Explicit gains used by the certificate

Suppose $R \ge M$, put $\varepsilon = 1/M$, and choose $b_0 \le S/n$. At most n columns have each derivative order. Their mean order is at least $(b_0R-1)/2$, and at most j_*R, where

$$j = 1/9 + \varepsilon, \qquad j_* = j - b_0/2 + b_0/M. \tag{24}$$

The latter is a conservative consequence of sorting the orders: the ith is at most $J-1-\lfloor (N-1-i)/n \rfloor$.

The b and higher-column terms in (22) give

$$G_{\mathrm{small}} = \sum_{p<Q_0/4} \log p \left\{ \sum_{\substack{e\ge 2\\ p^e<M}} \left(\frac{b_0-\varepsilon}{p^e} - 2\varepsilon \right)_+ + \left(\frac{t-\varepsilon}{p^2} - 3\varepsilon \right)_+ \right\}. \tag{25}$$

For the residual factor, define

$$A_p(t) = \begin{cases} 1/(2p^2) - t/(2p) + t^2/6, & pt \le 1, \\ 1/(6tp^3), & pt \ge 1. \end{cases}$$

Then the further gains are

$$G_{\mathrm{res}} = \sum_{p<Q_0/4} \log p \left(\frac{2A_p(t)}{p} - (2 + 2/p^3)\varepsilon \right)_+, \tag{26}$$

$$G_{\mathrm{coup}} = 2 \sum_{p<Q_0/4} \log p \left((1-1/p)\{A_p(t) - (1+1/p^2)\varepsilon - j_*\} - \varepsilon \right)_+. \tag{27}$$

Here is a direct integral justification. Put $x_r = r/R$ and $y_s = (S-1-s)/R$. We have $c/R \ge (x_r - py_s)_+/p^2 - 1/R$. On the cell $[x_r, x_r+1/R] \times [y_s, y_s+1/R]$, $(x-py)_+ \le (x_r - py_s)_+ + 1/R$. The average of $(x-py)_+/p^2$ over $[0,1]\times[0,S/R]$ equals $A_p(S/R)$, which decreases as S/R increases. Thus the normalized mean of c is at least $A_p(t) - (1+1/p^2)/M$. For (26), use the exact identity $\lfloor c/p \rfloor = \lfloor (r-pa)/p^3 \rfloor$. The same cell argument, now divided by p^3, bounds its normalized mean below by $A_p(t)/p - (1+1/p^3)/M$. The mean of a positive part is at least the positive part of its mean, so (23) and (24) give (27). All these terms are added to (20).

8 The factorial refinement uses the same smoothed sum

At a split prime, replace w^s by its falling factorial $(w)_s$. Since $\mathcal{O}_K/p^e\mathcal{O}_K \simeq (\mathbb{Z}/p^e\mathbb{Z})^2$, $p^{v_p(s!)}$ divides its value. The column factor $k!$ remains. At the ramified prime keep the column factorial only. These row changes are triangular, and therefore bound the same determinant.

Suppose a nonnegative nondecreasing function Φ is already known to satisfy $\mathcal{T}(e^x) \ge \Phi(x)$ throughout the band. The first-power part of the split row factorial contributes exactly $(S/R)\mathcal{T}_{\mathrm{split}}(S)$, by (13). It is therefore at least

$$(t-\varepsilon)\left(\Phi(\log(M(t-\varepsilon))) - \frac{\log Q_0}{Q_0} \right)_+.$$

For the columns put $K_0 = \lfloor b_0 R \rfloor$. The increasing chosen orders are at least $\lfloor i/n \rfloor$. Since $N/n \ge K_0$, the average of $\lfloor k/p \rfloor$ over these columns is at least its average for $0 \le k < K_0$. Using (13) again

gives a contribution at least $(K_0/R)\mathcal{T}(K_0)$. Hence a compatible total gain is

$$G_{\mathrm{fb}} = (b_0-\varepsilon)\Phi(\log(M(b_0-\varepsilon))) + (t-\varepsilon)\left(\Phi(\log(M(t-\varepsilon))) - \frac{\log Q_0}{Q_0}\right)_+. \tag{28}$$

There is no fractional-part approximation in this step. We start with $\Phi = 0$, compute a new lower bound, and repeat eight times. Every stage uses only the preceding stage, so the procedure is acyclic.

9 A joint bound for derivative orders and radii

We give the pointwise inequality underlying the archimedean estimate. Fix $\tau = 3/10$, and, for $0 \le k \le u \le 1$, put

$$F(u,k) = u\log u - (u-k)\log(u-k) - k, \qquad \phi(u) = \tau(u\log u - u).$$

At zero use the continuous values. For $0 < a \le 1$, define $\psi(k,a) = \max_{k\le u\le 1}\{(u-k)\log a + F(u,k) - \phi(u)\}$. The maximized function is strictly concave. If $a \le 1-k$, its critical point satisfies $u - k = au^{1-\tau}$, and

$$\psi(k,a) = \tau u + (1-\tau)k\log u - k.$$

If $a \ge 1-k$, the maximum is at 1 and

$$\psi(k,a) = (1-k)\log\frac{a}{1-k} - k + \tau.$$

Differentiation shows that ψ decreases in k, increases in a, and is 1-Lipschitz in a. In particular, for $k \le j$,

$$\psi(k,a) + B \ge 0, \qquad B = (1-\tau)j(1-\log j).$$

Let $I(k) = \int_0^1 2a\psi(k,a)\,da$ and $K(j) = \int_0^j I(k)\,dk$. We have $I(0) = 2\tau^2/(2\tau+1)$. Changing variables from a to the critical point gives

$$I'(k) = -(1-\tau)\int_k^1 (u-k)^2 u^{2\tau-3}\,du + \log(1-k) + k - k^2/2.$$

Thus, writing $A = 2\tau$,

$$K(j) = j\frac{2\tau^2}{2\tau+1} - (1-\tau)\left\{\frac{j^2}{2A} - \frac{j^3}{3(A-1)} + \frac{j^4}{12(A-2)} - \frac{2j^{A+2}}{A(A-1)(A-2)(A+1)(A+2)}\right\} + \frac{j}{2} - \frac{3j^2}{4} + \frac{(1-j)^2}{2}\log(1-j) + \frac{j^3}{6} - \frac{j^4}{24}. \tag{29}$$

For an actual row and derivative order, take $u = (r+1)/R$ and $k' = k/R$. Increasing-log integration gives $\log(r)_k \le k\log R + RF(u,k')$. Clip the normalized radius to $a = \max(1,|z|)/T$. A nonzero entry satisfies

$$\log|\partial_z^k(z^r w^s)| \le (r+s)\log T + k\log(R/T) + R\{\phi(u) + \psi(k',a)\} + s\log a - \log a.$$

The final term is at most $\log T$. Since ϕ decreases, its right Riemann sum gives a normalized row contribution at most $-3\tau/2$ to $\log\Delta/(NR)$.

For uniform estimates set $\varepsilon = 1/M$,

$$r_* = \sqrt{CMv_+/\pi}, \quad e \ge \rho_+/r_*, \quad \nu = C(1+2e)^2, \quad b_0 \le (t-\varepsilon)/\nu, \quad g = \tfrac{1}{2}\log(C/(2\pi)) + \log(1+e). \tag{30}$$

Here $v_+ = \sqrt{Q_1}/2$, $\rho_+ = \sqrt{1+v_+^2}/2$. Then $n \le \nu R$, $b_0 \le S/n$, and $\log T \le \frac{1}{2}\log R + \frac{1}{4}\log q + g$. The monotonicity of $\rho/\sqrt{v} = \frac{1}{2}\sqrt{v+v^{-1}}$ justifies use of the upper band endpoint.

Lattice-cell integration, using nonnegativity and the Lipschitz property, gives for each order

$$\sum_{|z|\le T} \{\psi(k/R, a_z) + B\} \le \nu R \left\{ I(k/R) + B + \frac{5e}{3(1+e)} \right\}.$$

Indeed, cells lie in the disk of radius $T+\rho$. Its normalized radial argument is $((1+2e)a+e)/(1+e)$, whose difference from a is $e(1+a)/(1+e)$. Integrating against $2a\,da$ gives the displayed error. Clipping at 1 only improves the estimate; $\rho \ge 1$ also covers the clipped origin.

We may fill in unchosen columns before subtracting NB. Since I decreases and $I + B \ge 0$, the order sum is at most

$$R\left\{ K(j) + j\left(B + \frac{5e}{3(1+e)} \right) \right\} + I(0) + B.$$

It follows that the normalized joint contribution is at most

$$D_* = -\frac{3\tau}{2} + \frac{2\nu}{t-\varepsilon}\left\{ K(j) + j\left(B + \frac{5e}{3(1+e)} \right) + \varepsilon(I(0)+B) \right\} - 2B. \tag{31}$$

9.1 The unused s-radius contribution

At least $\lambda(a)N$ chosen columns have radius at most aT, where

$$\lambda(a) = \left(1 - \frac{jC}{t-\varepsilon}\{(1+2e)^2 - \max(a(1+e)-e, 0)^2\} \right)_+.$$

Among that many distinct rows the normalized sum of the s's is at least $((t-\varepsilon)\lambda(a)^2 - \varepsilon)/2$. Put

$$\delta = \frac{jC}{t-\varepsilon}\left\{ (1+2e)^2 + \frac{e}{2+e} \right\} - 1.$$

Since $\max(a(1+e)-e, 0)^2 \ge a^2 - e/(2+e)$ and $jC/(t-\varepsilon) \ge 1$, we have $\lambda(a) \ge (a^2-\delta)_+$. The certificate checks $0 < \delta < 1$ and $\delta \ge e$, so $\sqrt{\delta} \ge 1/T$. Integrating from $\sqrt{\delta}$ to 1 gives the additional saving

$$G_s = \left((t-\varepsilon)\left\{ \frac{1}{4} - \delta + \frac{3\delta^2}{4} - \frac{\delta^2}{2}\log\delta \right\} + \frac{\varepsilon}{2}\log\delta \right)_+. \tag{32}$$

There are at most $N!$ terms in the determinant. Using $\log N! \le N\log N$, (24), and $1 < T < R$, we obtain

$$\frac{\log\Delta}{NR} \le (1-\alpha)\log R + \frac{\beta}{4}\log q + \beta g + D_* + \frac{2\log M}{M} - G_s + \frac{2\log(tM^2)}{M}, \tag{33}$$

where $\alpha = (1-t)/2 - j_*$, $\beta = 1 + t - 2j_*$. The functions $(\log R)/R$ and $\log(tR^2)/R$ decrease on the certified ranges. Notice that the joint estimate replaces the former separate derivative and radius estimates; incompatible savings are not added.

10 The certified affine bounds and finite iteration

Combine (20), (6), (25)–(27), (28), and (33). This gives

$$\mathcal{T}(R) \geq \alpha \log R - \frac{\beta}{4} \log q - \mathcal{K}, \tag{34}$$

where the fully explicit constant is

$$\begin{aligned}\mathcal{K} = {} & a_H - W_H + c(\log M) + \frac{B_H - K_2}{3(t-\varepsilon)} + \beta g + D_* + \frac{2\log M}{M} - G_s \\ & + \frac{2\theta_H}{M} + \frac{\log M}{(t-\varepsilon)M} + \frac{2\log(tM^2)}{M} + 1.334 + \frac{1}{2\log M} \\ & - G_{\text{floor}} - G_{\text{small}} - G_{\text{res}} - G_{\text{coup}} - G_{\text{fb}}.\end{aligned}$$

For every $U \in [2^a, 2^b]$, the auxiliary prime satisfies $U < R \leq \kappa U$. Monotonicity of $\mathcal{T}$ and $\alpha > 0$ therefore transfer (34) to $X = \kappa U$, with the additional intercept cost $\alpha \log \kappa$.

The certificate rounds slopes down and intercept costs up, obtaining

$$\mathcal{T}(e^x) \geq A_i x - D_i - \gamma \log(q/Q_0), \qquad \gamma = 3/10,$$

on the indicated finite interval. It checks $\beta/4 < \gamma$. At a finite endpoint the endpoint value remains a lower bound at all later x. The maximum with zero is nondecreasing. For each of eight refinement stages, use this preceding maximum minus $\gamma \log(Q_1/Q_0)$, clipped at zero, as Φ in (28). The initial stage uses $\Phi = 0$.

Write $L(x)$ for a certified piecewise affine selection from the final bounds at Q_0, including zero and frozen endpoint values. Then throughout the band,

$$\mathcal{T}(e^x) \geq L(x) - \gamma \log(q/Q_0). \tag{35}$$

The exact parameter and output files record all lines, stages, and selected segments. No optimizer is part of the certificate.

11 Positive kernels and the quadratic norm series

Choose positive rational d, e and a nonzero integer polynomial P, and normalize

$$h(x) = Z^{-1} e^{-dx} P(e^{-ex})^2 \quad (x \geq 0), \qquad Z = \int_0^\infty e^{-dx} P(e^{-ex})^2 \, dx.$$

Then $h \geq 0$ and $\int h = 1$. Expand $h(x) = \sum_i c_i d_i e^{-d_i x}$, with $d_i = d + ie$. All c_i and Z are exact rationals. Integrating (35) gives

$$\int_0^\infty h(x) \mathcal{T}(e^x) \, dx \geq \sum_i c_i \int_0^\infty d_i e^{-d_i x} L(x) \, dx - \gamma \log(q/Q_0). \tag{36}$$

One and the same L is used in every signed term.

Define

$$W(y) = \int_y^\infty h(x) f(e^{y-x}) \, dx.$$

This is nonnegative, nonincreasing, and at most 1. Tonelli's theorem and (13) show that

$$W(y) = \sum_i c_i F(d_i) e^{-d_i y}, \qquad F(d) = \int_0^1 d\, u^{d-1} f(u)\, du = \frac{2d\zeta(1+d)}{(1+d)(2+d)}. \tag{37}$$

The right side follows by integrating each $2d\, u^d (1-mu)_+$ from 0 to $1/m$. Thus the left side of (36) is $\sum_{p \text{ noninert}} (\log p)/p\, W(\log p)$.

For $s > 1$, put

$$\mathcal{A}_q(s) = \sum_{b \ge 1} \sum_{a \in \mathbb{Z}} Q_q(a,b)^{-s}, \qquad B_q(s) = 1 + \mathcal{A}_q(s)/\zeta(2s).$$

Class number one and the unit group give $\zeta_K(s) = \zeta(2s) + \mathcal{A}_q(s)$. Its Euler product implies

$$-\frac{B_q'(s)}{2B_q(s)} = \sum_{p \text{ split}} \sum_{\substack{m \ge 1 \\ m \text{ odd}}} \frac{\log p}{p^{ms}} + \frac{\log q}{2} \sum_{m \ge 1} (-1)^{m-1} q^{-ms}.$$

Multiply by the signed coefficients $c_i F(d_i)$ and sum at $s_i = 1 + d_i$. Since $W \ge 0$, the discarded higher odd powers have nonnegative weights. Adding $\frac{1}{2} \log q \sum_{m \ge 1} q^{-m} W(m \log q)$ also covers the ramified first-power term. It is at most

$$\frac{\log Q_0}{2(Q_0 - 1)}. \tag{38}$$

12 A small explicit norm-series error

Set

$$C(s) = \sqrt{\pi} \frac{\Gamma(s - 1/2)}{\Gamma(s)}, \quad g(s) = \psi(s) - \psi(s - 1/2), \quad \mathcal{F}_q(s) = C(s) v^{1-2s} \zeta(2s-1).$$

Poisson summation in the real coordinate gives a particularly small error for $\mathcal{A}_q - \mathcal{F}_q$. We supply a bound suitable for the signed kernel coefficients.

Choose any rational $0 < a_0 < v_-$. For $f_{s,t}(x) = (x^2 + t^2)^{-s}$, shift the Fourier contour by $y = t - a_0$, where $t = bv$. Since

$$|(x + iy)^2 + t^2| \ge x^2 + a_0^2,$$

the nonzero Fourier coefficients are bounded by $C(s) a_0^{1-2s} e^{-2\pi n (t - a_0)}$. Summing both signs and all $b, n \ge 1$, put

$$z = e^{-2\pi v_-}, \quad y_0 = e^{-2\pi(v_- - a_0)}, \qquad E_A = \frac{2C(s) a_0^{1-2s} y_0}{(1-z)(1-y_0)}.$$

Then $|\mathcal{A}_q(s) - \mathcal{F}_q(s)| \le E_A$ uniformly in the band.

For the s-derivative, the complex argument of $(x + iy)^2 + t^2$ has absolute value at most $\pi/2$, and its logarithm is bounded in modulus by

$$\log(1 + x^2/a_0^2) + 2\log(2t) + 2|\log a_0| + \pi/2.$$

The integral of the first term against $(x^2 + a_0^2)^{-s}$ is $C(s) a_0^{1-2s} g(s)$. Using $\log b \le b - 1$ in the remaining geometric sum therefore gives

$$|\mathcal{A}_q'(s) - \mathcal{F}_q'(s)| \le E_N, \quad E_N = E_A \left\{ 2\log(2v_+) + 2|\log a_0| + g(s) + \frac{\pi}{2} + \frac{2z}{1-z} \right\}. \tag{39}$$

The contour shift is justified by decay at infinity for $s > 1$; the same decay with an extra logarithm justifies differentiation.

Put $d = s - 1$, $z_2 = \zeta(2s)$, $c_s = -2\zeta'(2s)/\zeta(2s)$, and

$$\kappa_d = g(s) - 2\frac{\zeta'(2s-1)}{\zeta(2s-1)} - c_s.$$

The main upper term is

$$H_q(d) = \frac{\mathcal{F}_q(s)\{\log(q/4) + \kappa_d\}}{2\{z_2 + \mathcal{F}_q(s)\}}. \tag{40}$$

The certificate checks that $H_q(d) > 0$ and decreases with q for each exponent used. Since $z_2 > 1$, the logarithmic-derivative error is at most

$$E_H(d) = \frac{E_N + (c_s + 2H_{Q_0}(d))E_A}{2(1 - E_A)}. \tag{41}$$

Consequently a uniform upper bound for the weighted noninert sum is

$$U(q) = \sum_i c_i F(d_i) H_q(d_i) + \sum_i |c_i| F(d_i) E_H(d_i) + \frac{\log Q_0}{2(Q_0 - 1)}. \tag{42}$$

The error is bounded after all coefficient magnitudes are included.

13 Exact certification and completion of the proof

The certificate uses the five bands

$$[570, 580], \quad [580, 600], \quad [600, 700], \quad [700, 1050], \quad [1050, 2000].$$

The first uses a polynomial of degree 16 with $d = 3/10, e = 1/25$; the others use degree 12 with $d = 2/5, e = 1/20$. The integer polynomial coefficients, auxiliary-prime rows, and eight-stage refinement schedule are fixed in `parameters.json`.

For completeness, the dependence on q is certified over each entire band. Write $Y = \log(q/Q_0)$, $a_i = 1/2 + d_i$, and $r_i = \mathcal{F}_{Q_0}(s_i)/\zeta(2s_i)$. Expand

$$H_q(d_i) = \frac{1}{2}(\log(Q_0/4) + Y + \kappa_{d_i}) \sum_{\ell \geq 1} (-1)^{\ell+1} r_i^\ell e^{-\ell a_i Y}.$$

The certificate bounds the differentiated geometric tail explicitly. For each retained ℓ, the signed finite sum is evaluated as a polynomial in $1 - e^{-\ell e Y}$, with its coefficients computed by exact interval arithmetic. This avoids naive interval cancellation. Cells of width $1/1000$ cover the entire Y-interval. On every cell the derivative of the main sum in (42) is strictly less than $-\gamma$. The two error bounds in that equation are uniform constants, so they require no differentiation.

Band	Lower bound	Upper bound	Gap >	Norm error <
$[570, 580]$	10.975413	10.965024	0.010389	7.652×10^{-5}
$[580, 600]$	9.566392	9.544454	0.021938	1.969×10^{-10}
$[600, 700]$	9.755174	9.682808	0.072367	4.675×10^{-11}
$[700, 1050]$	9.890348	9.507108	0.383240	6.739×10^{-14}
$[1050, 2000]$	12.460477	11.013702	1.446776	3.672×10^{-22}

All displayed bounds are rounded outward. The upper bound includes both the norm-series error and the ramified-prime allowance. The exact certificate endpoints are recorded in the JSON output.

At $q = Q_0$, the exact lower endpoint of (36) is strictly greater than the exact upper endpoint of (42). Since the latter decreases faster than $-\gamma \log(q/Q_0)$, (35) proves a contradiction throughout the band. The five bands exclude $[570, 2000]$; the audited v5 implication excludes $q \geq 2000$, proving Theorem 1.

The numerical proof uses integer and rational intervals throughout. Logarithms use a convergent positive series after range reduction; exponentials use a Taylor series with an explicit tail; square roots use integer square roots. Zeta values and derivatives use Euler–Maclaurin with the periodic-Bernoulli remainder, while gamma and digamma use positive-real Stirling expansions with remainder bounds. These are evaluations in $\Re s > 1$, not zero-location inputs. Acceptance never depends on a floating-point optimizer.

Run, without Python's `-O` option,

```
python3 verify_v6.py
python3 verify_structures.py
python3 verify_finite.py
```

The first command certifies the inequalities and writes exact endpoints to JSON. The second checks finite instances of the new algebraic identities and divisibility statements; these tests do not replace their proofs. The third is a separate finite check below the analytic cutoff and is not used to establish that cutoff.

13.1 Numerical remainder bounds

Here are the remainder formulas used in the certificate; the integer parameters in this paragraph are numerical evaluation parameters, independent of the determinant dimensions. For $s > 1$, Euler–Maclaurin with $N_0 = 64$, $m = 18$ gives

$$\begin{aligned}
\zeta(s) &= \sum_{n<N_0} n^{-s} + \frac{N_0^{1-s}}{s-1} + \frac{1}{2}N_0^{-s} + \sum_{k=1}^{m} \frac{B_{2k}(s)_{2k-1}}{(2k)!} N_0^{1-s-2k} + \mathcal{R}, \\
|\mathcal{R}| &\leq E = \frac{|B_{2m}|(s)_{2m}}{(2m)!(s+2m-1)} N_0^{1-s-2m}, \\
|\mathcal{R}'| &\leq E \left\{ \log N_0 + \sum_{j=0}^{2m-1} \frac{1}{s+j} + \frac{1}{s+2m-1} \right\}.
\end{aligned}$$

Here $(s)_j$ denotes the rising factorial. These bounds follow from the periodic-Bernoulli integral and its derivative. For log gamma and digamma, shift to $z = x + 64$, retain the terms $1 \leq k < 20$ in the positive-real Stirling expansions, and use the first omitted term as an absolute remainder bound [12]:

$$|R_{\log\Gamma}| \leq \frac{|B_{40}|}{40 \cdot 39\, z^{39}}, \qquad |R_{\psi}| \leq \frac{|B_{40}|}{40 z^{40}}.$$

The recurrence formulas then recover the values at x.

After range reduction to $1 \leq y \leq 2$, logarithms use 80 terms of $2\sum_{k\geq 0} t^{2k+1}/(2k+1)$, where $t = (y-1)/(y+1)$, and tail $2t^{161}/(161(1-t^2))$. Exponentials are reduced to $0 \leq x \leq 1$; the series through degree 80 has positive tail at most $82/(81 \cdot 81!)$. Repeated squaring and reciprocation undo the reduction. Machin's formula with 60 arctangent terms encloses π. Elementary interval operations round outward to 65 decimal places; square roots use integer bounds.

For the differentiated geometric series above, the tail after L terms is at most

$$\frac{1}{2}\sum_i |c_i|F(d_i)r_i^{L+1}\left\{\frac{1}{1-r_i}+a_iE_*\left(\frac{L+1}{1-r_i}+\frac{r_i}{(1-r_i)^2}\right)\right\}, \qquad E_*=\max_i\{\log(Q_1/4)+\kappa_{d_i}\}.$$

The certificate increases L until this bound is below 10^{-10}, then includes it in every derivative cell test.

14 Directions tested and the stopping point

The research continued beyond the first successful lighter-method bound. The retained improvements include disk interpolation, the prime itself as monomial width, a mixed ideal sieve, higher Frobenius factors, the smoothed valuation sum, direct finite factorial refinement, the joint radius/order inequality, and optimized positive kernels.

Other routes were examined but are not inputs to this proof. Weighted monomial regions suggested by Kerr lose the exact packing of the degrees $r+p_0s$ used here. Numerical tests with disk orthogonal polynomials gave only a small further improvement over the joint monomial estimate. Finer transport bounds and more elaborate kernels offered diminishing returns, with increasingly difficult signed-sum error control. The sieve work of Gaudet [11] was also considered; its additional class-group analytic inputs and unspecified numerical exponent did not supply a useful replacement for the present explicit argument. The quartic-field/Masser and quadratic zero-free approaches were not imported.

The cutoff 570 is a practical stopping point for this research pass, not an optimality theorem about these methods. The package records the final proof and certificate as one v6 result, and preserves the distinction between exact arithmetic, mathematical audit, and formal verification.

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